

2023 AMC 10B Problem 25 - Solution

Review the full statement and step-by-step solution for 2023 AMC 10B Problem 25. Great practice for AMC 10, AMC 12, AIME, and other math contests

2023 AMC 10B Problem 25

Problem:

A regular pentagon with area $1 + \sqrt{5}$ is printed on paper and cut out. All five vertices are folded to the center of the pentagon, creating a smaller pentagon. What is the area of the new pentagon?

Answer Choices:

A. $4 - \sqrt{5}$

B. $\sqrt{5} - 1$

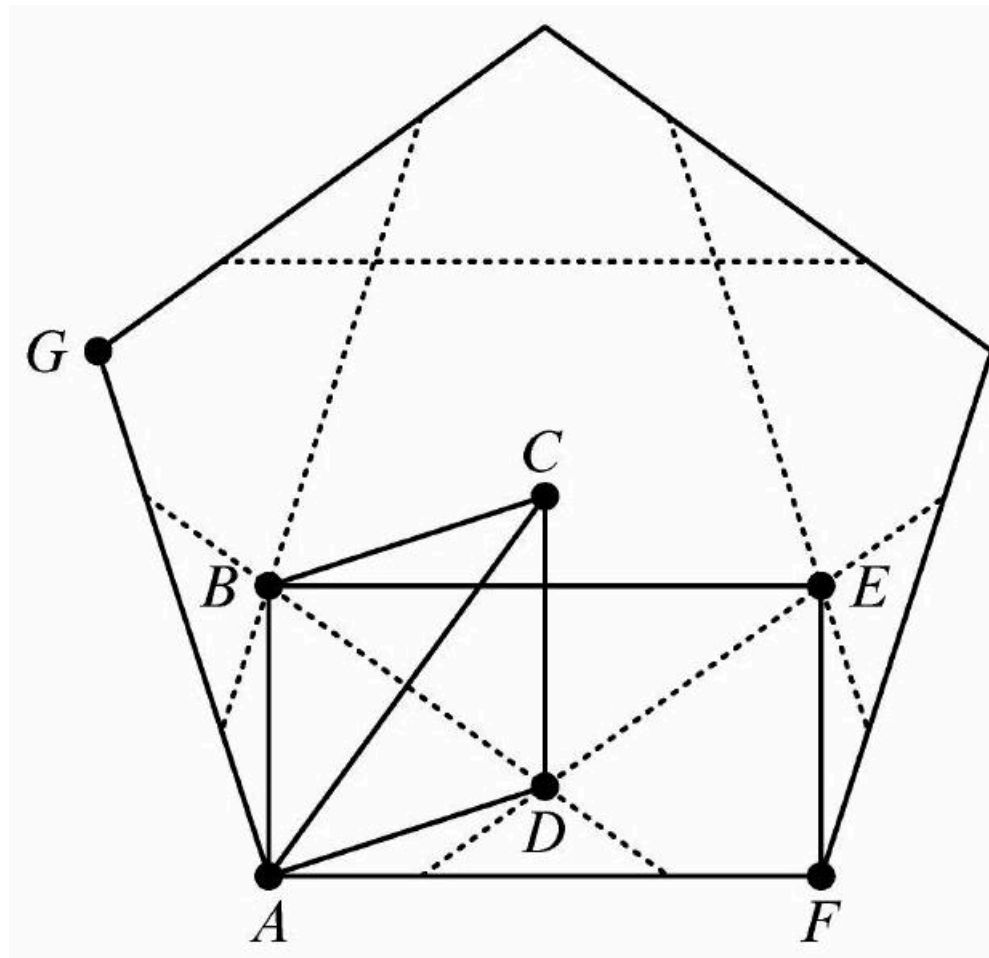
C. $8 - 3\sqrt{5}$

D. $\frac{1 + \sqrt{5}}{2}$

E. $\frac{2 + \sqrt{5}}{3}$

Solution:

The folds are the perpendicular bisectors of the segments connecting the vertices of the pentagon to its center. Let C be the center of the pentagon, let G , A , and F be three consecutive vertices in order, let D be the intersection of the fold lines for vertices A and F , let B be the intersection of the fold lines for vertices A and G , and let E be the other intersection of fold lines on the fold line for F . Draw quadrilaterals $ABCD$ and $ABEF$, as shown.



Let S be the area of the small pentagon. Because the two pentagons are similar, the ratio of their areas is the square of the ratio of their side lengths. Therefore

$$\frac{S}{1 + \sqrt{5}} = \left(\frac{BD}{AF} \right)^2.$$

By symmetry, $ABCD$ must be a rhombus, and $\overline{AC}$ bisects $\angle BAD$ (and $\angle GAF$). Then

$$\angle BAD = \angle BCD = \frac{360^\circ}{5} = 72^\circ,$$

so $\angle BAC = \frac{72^\circ}{2} = 36^\circ$. Also, $\angle GAF = \frac{3 \cdot 180^\circ}{5} = 108^\circ$, so $\angle CAF = \frac{108^\circ}{2} = 54^\circ$. Lastly,

$$\angle BAF = \angle BAC + \angle CAF = 36^\circ + 54^\circ = 90^\circ.$$

Therefore quadrilateral $ABEF$ is a rectangle and $BE = AF$. It is known (and proved below) that the ratio of a diagonal of a regular pentagon to its side is $\frac{1+\sqrt{5}}{2}$ (φ , the golden ratio). Therefore

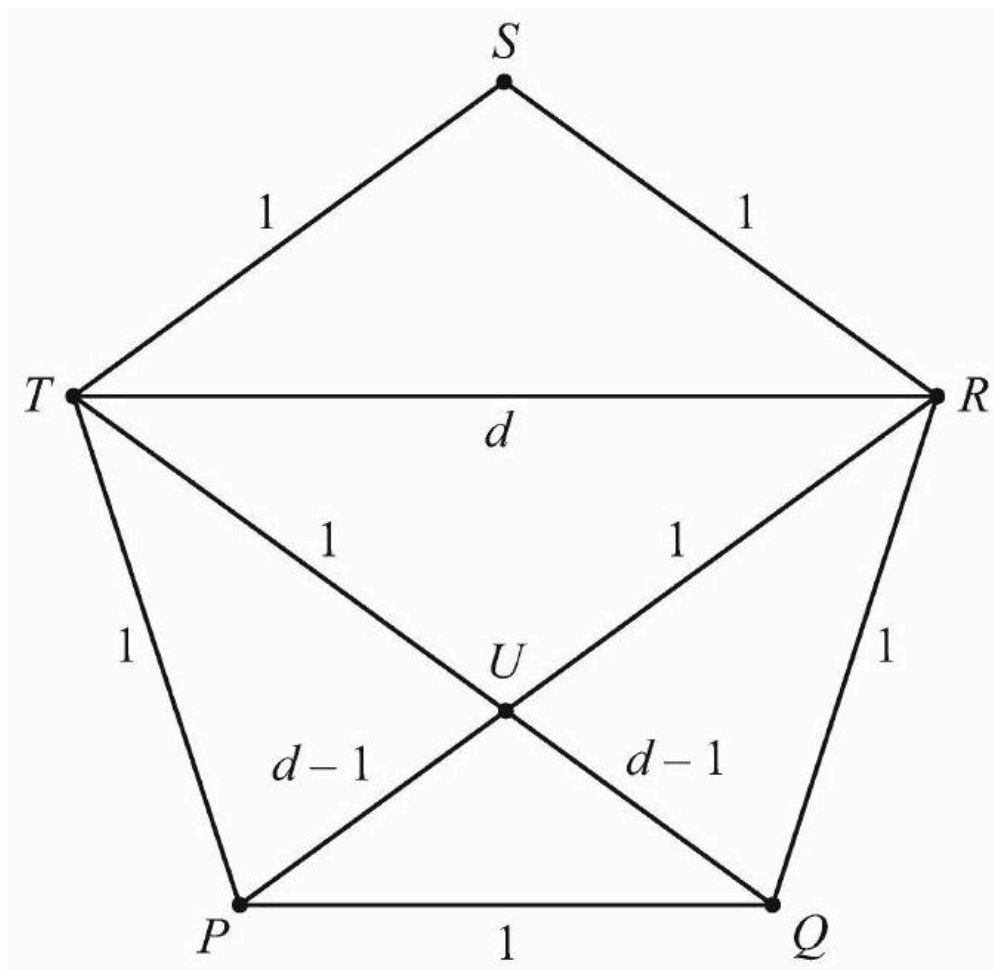
$$\frac{BD}{AF} = \frac{BD}{BE} = \frac{1}{\varphi} = \frac{2}{1 + \sqrt{5}},$$

and

$$\frac{S}{1 + \sqrt{5}} = \left(\frac{BD}{AF} \right)^2 = \left(\frac{2}{1 + \sqrt{5}} \right)^2.$$

The requested area is $S = \frac{2^2}{1+\sqrt{5}} = \text{(B)} \boxed{\sqrt{5} - 1}$.

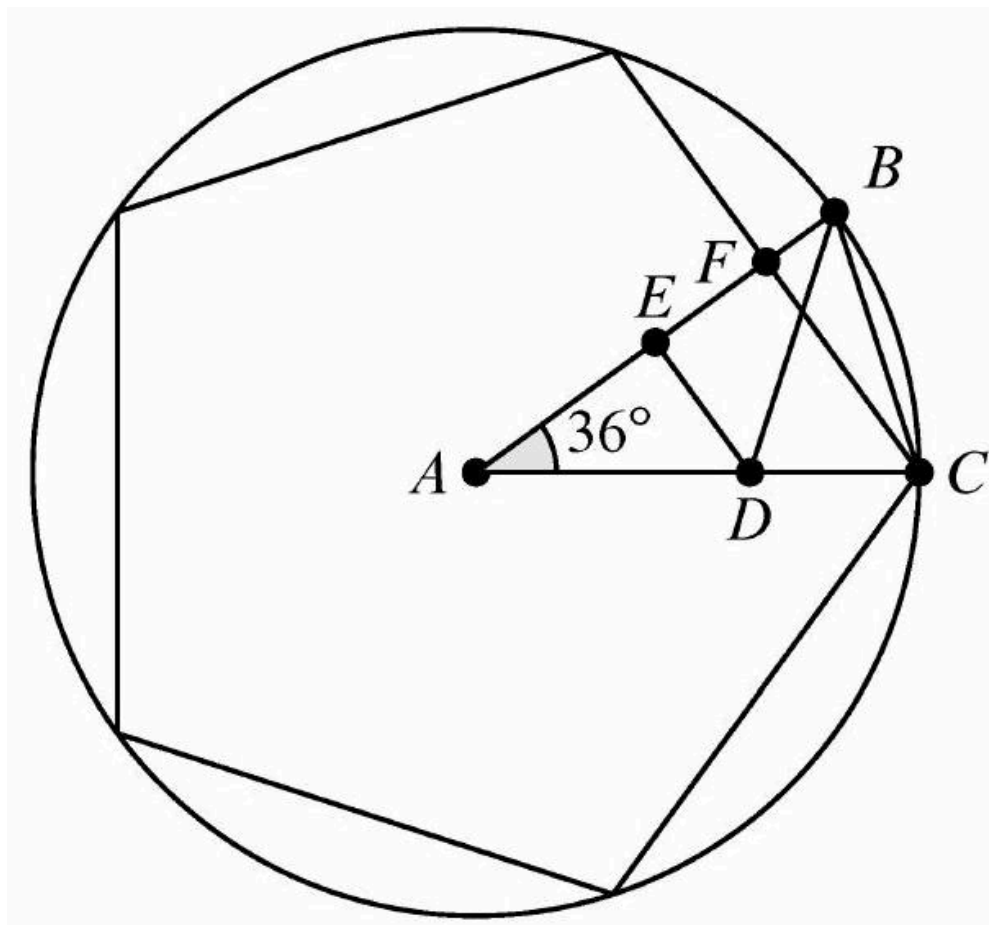
To prove that the diagonal to side ratio for a regular pentagon equals $\frac{1+\sqrt{5}}{2}$, let regular pentagon $PQRST$ have sides of length 1 and diagonals of length d , and let U be the point of intersection of the line segments $\overline{PR}$ and $\overline{QT}$.



Because each diagonal is parallel to a side, parallelogram $TURS$ is formed with sides of length 1, leaving segments $\overline{PU}$ and $\overline{QU}$ to have length $d - 1$. Because $PQRT$ is a trapezoid, $\triangle RTU$ and $\triangle PQU$ are similar, and $\frac{d}{1} = \frac{1}{d-1}$. This implies that $d^2 - d - 1 = 0$, and the quadratic formula gives the positive solution $d = \frac{1+\sqrt{5}}{2} = \varphi$.

OR

Inscribe the regular pentagon in a circle of radius r . Let A be the center of the circle, let C be a vertex of the pentagon, and let B be the midpoint of the minor arc from C to one of the vertices adjacent to C . Let D be the intersection of the bisector of $\angle ABC$ with $\overline{AC}$. Let E be the midpoint of $\overline{AB}$, and let F be the intersection of $\overline{AB}$ with the pentagon. Note that AF is the center-to-side distance of the pentagon. See the figure.



The linear dimensions are reduced by the factor $f = \frac{r}{2 \cdot AF}$, and the area is reduced by the factor f^2 . The scale factor f can be evaluated using similar triangles. Triangle $\triangle ABC$ is isosceles with angles $36^\circ, 72^\circ, 72^\circ$ and sides $AB = AC = r$. Segment $\overline{BD}$ bisects $\angle ABC$ creating isosceles triangles $\triangle ABD$ with two 36° angles and $\triangle BDC$ with two 72° angles. Choose units so that $1 = BC = BD = AD$. Notice that $\triangle BDC \sim \triangle ABC$ and is smaller by a factor of r . Because

$AD + DC = AC$, it follows that $1 + \frac{1}{r} = r$. Because $r > 1$, this simplifies to $r = \frac{1+\sqrt{5}}{2}$. Now

$$f = \frac{AC}{2 \cdot AF} = \frac{AD}{2 \cdot AE} = \frac{1}{r} = \frac{2}{1 + \sqrt{5}}.$$

The area of the smaller pentagon is

$$f^2(1 + \sqrt{5}) = \frac{4(1 + \sqrt{5})}{(1 + \sqrt{5})^2} = (\text{B}) \boxed{\sqrt{5} - 1}$$

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